

Conclusion: ~~Axiom~~ All compound propositions are logically equivalent to a compound proposition in disjunctive normal form $\square$

[Note: The written soln. should be better]

A collection of logical operators is called functionally complete if every compound proposition is logically equivalent to a compound proposition involving only these logical operators.

47) Show that $\neg$, $\wedge$ and $\vee$ form a functionally complete collection of logical operators.

[Soln:] From Exercise 46, we know ~~for~~ all compound propositions are logically equivalent to a compound proposition in disjunctive normal form. From the defn. of disjunctive normal form, we know the compound proposition is only ~~to~~ composed of the logical operators $\neg$, $\wedge$ and $\vee$.

$\therefore$ The three operators $\neg$, $\wedge$ and $\vee$ are functionally complete $\square$

48) Show that $\neg$ and $\wedge$ form a functionally complete collection of logical operators.

[Soln:] Let, ~~$(P_1 \vee P_2 \vee P_3 \vee \dots \vee P_n)$~~ be a compound proposition in disjunctive normal form where $\forall i, 1 \leq i \leq n, P_i$ ~~can~~ is a conjunction comprising of the m propositional variables $q_1, q_2, \dots, q_m$ or their negations. From Ex-46, we know all compound propositions are logically equivalent to a proposition in DNF. From Double negation law,

$$\begin{aligned} (P_1 \vee P_2 \vee P_3 \vee \dots \vee P_n) &\equiv \neg(\neg(P_1 \vee P_2 \vee P_3 \vee \dots \vee P_n)) \\ &\equiv \neg(\neg P_1 \wedge \neg P_2 \wedge \neg P_3 \wedge \dots \wedge \neg P_n) \quad (\text{By 2nd De Morgan's Law and Mathematical Induction, the full proof of which is in Chapter 5.}) \end{aligned}$$

$\therefore \neg$ and $\wedge$ form a functionally complete collection of Logical operators

$\square$